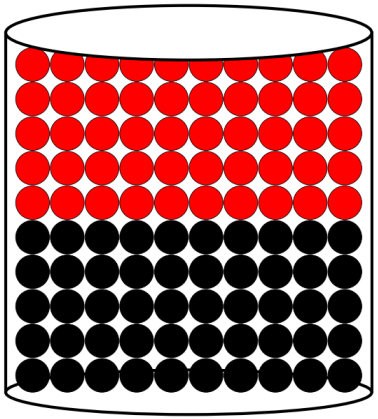
Are Ambiguity Attitudes Transferable? A Test Using DOSE

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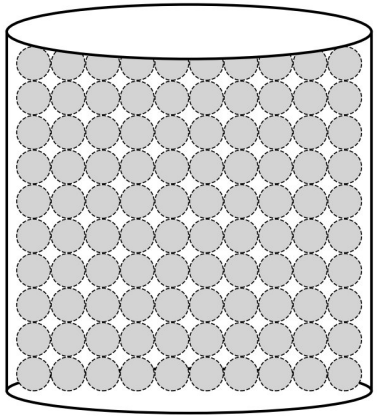
Risk vs. Ambiguity

Risk



- Outcome probabilities are known.
- Preference under risk: risk attitude.

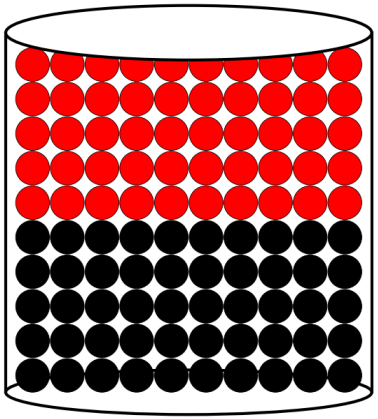
Ambiguity



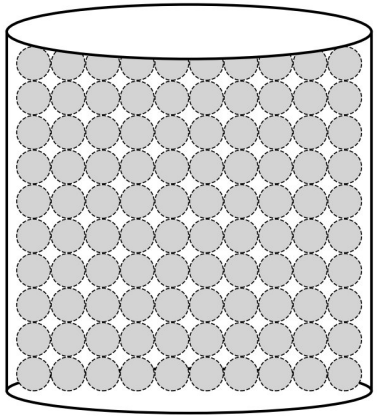
- Outcome probabilities are unknown.
- Additional preference beyond risk attitude: ambiguity attitude.

Risk vs. Ambiguity

Risk



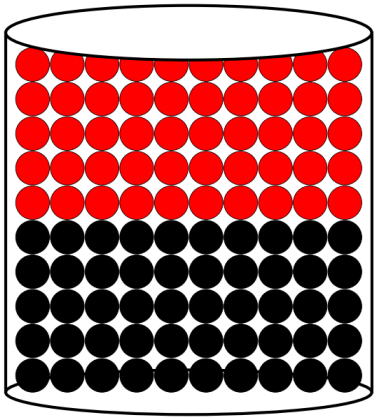
Ambiguity



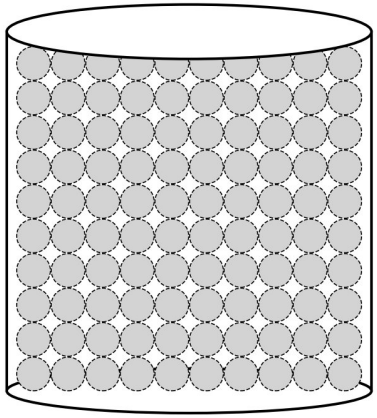
- **Bet 1:** Choose an urn. One ball is drawn from that urn. Win $\$x$ if the ball is red.
  - Most people choose the risky urn.
  - Without ambiguity attitudes, this suggests $P_A(\text{red}) < 0.5$.

Risk vs. Ambiguity

Risk



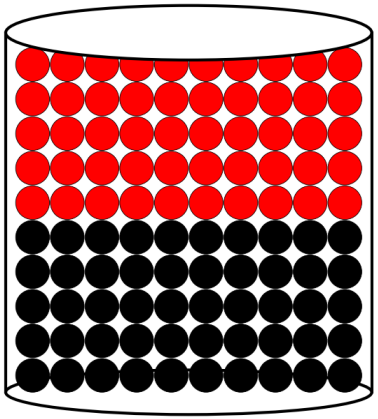
Ambiguity



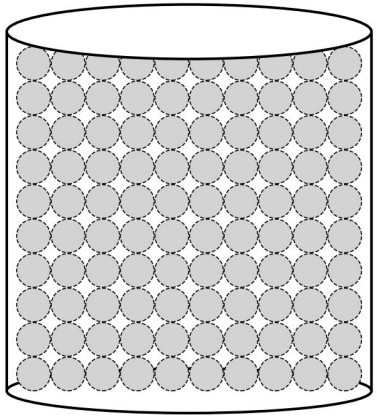
- **Bet 2:** Choose an urn. One ball is drawn from that urn. Win $\$x$ if the ball is black.
  - Most people choose the risky urn again.
  - Without ambiguity attitudes, this suggests $P_A(\text{red}) > 0.5$.

Risk vs. Ambiguity

Risk



Ambiguity



For decision makers who prefer the risky urn in both bets, without ambiguity attitudes:

$$P_A(\text{red}) < 0.5 \quad \text{and} \quad P_A(\text{red}) > 0.5$$

No single subjective belief can explain choosing the risky urn in both bets.
$\Rightarrow$ **Additional preference beyond risk attitude: ambiguity attitude.**

Do Ambiguity Attitudes Transfer to New Situations?

- Ambiguity attitudes matter for important economic decisions:
  - technology adoption (Bryan, 2019),
  - insurance pricing (Cabantous, 2007),
  - portfolio choice (Bianchi & Tallon, 2019),
  - stock market participation (Dimmock et al., 2016a),
  - and several others.
- Models that incorporate ambiguity attitudes could be promising tools for predicting these decisions.
- We often estimate ambiguity attitudes from choices made in one situation, but need to predict behavior when the situation changes.
- Do those estimates still have predictive value in new situations?

Should Ambiguity Attitudes Inform Policy Predictions?

A central bank wants to predict the effects of a proposed policy.



Situation 1 (Historical Data)	Situation 2 (Prediction Needed)
Households' ambiguity attitudes are estimated from their portfolio choices under existing market conditions.	A proposed policy would change market conditions, and households might adjust their portfolios accordingly.

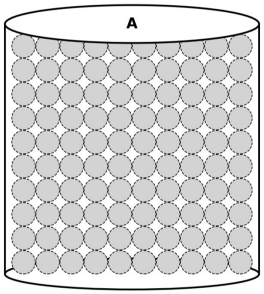
- Do estimated ambiguity attitudes remain useful for predicting behavior when situations change?
- If so, which ambiguity model predicts best?

Why Not Use Policy Experiments or Observational Data?

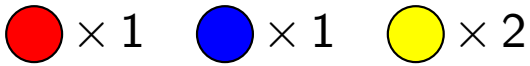
1. Policy experiments
   - No central bank would let us run this experiment.
   - We would need to implement different central-bank policies in two identical countries under the same conditions.
2. Observational data
   - Models that incorporate ambiguity attitudes may fit historical data better simply because they are more flexible.
   - It is difficult to determine whether observed behavior is actually driven by ambiguity attitudes.

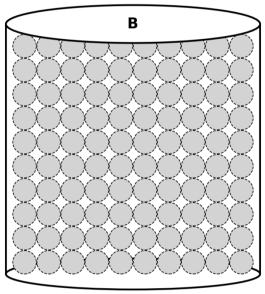
Our approach: Design a controlled experiment that allows us to cleanly isolate the role of ambiguity attitudes.

How Do We Test Whether Ambiguity Attitudes Transfer?

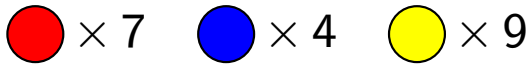


Situation 1:
4 draws from Box A



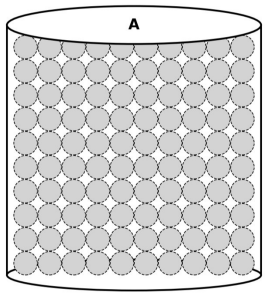


Situation 2:
20 draws from Box B

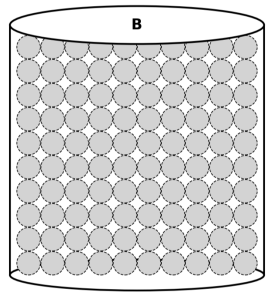


The two situations share the same decision context but differ in the information available.

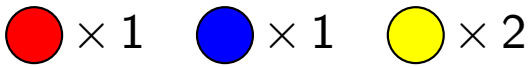
How Do We Test Whether Ambiguity Attitudes Transfer?



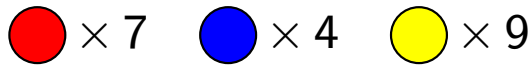
Do estimated ambiguity attitudes transfer?



Situation 1:
4 draws from Box A



Situation 2:
20 draws from Box B



We can cleanly attribute behavior to two main components:

- Beliefs: Can change with the available information.
- Ambiguity attitudes: Preferences toward ambiguity.

What we do:
For each model, estimate ambiguity-attitude parameters in Situation 1 and test whether they retain predictive value in Situation 2.

Beyond Practical Relevance: A Stronger Test of Models

If we compare models within a situation:

- Beliefs and ambiguity attitudes remain fixed.
- A model can predict choices well even if it attributes belief-driven behavior to ambiguity attitudes.

Instead, if we compare models across situations:

- Beliefs change across situations, while estimated ambiguity-attitude parameters are held fixed.
- Such misattribution can be revealed when testing predictions in a new situation.

$\rightarrow$ Cross-situation comparisons provide a stronger test of whether models correctly capture the roles of beliefs and ambiguity attitudes.

This Paper

- We use an online experiment to compare the out-of-sample predictive performance of three leading models of ambiguity:
  - Source Theory (Abdellaoui et al., 2011; Baillon et al., 2025)
  - Smooth Ambiguity (Klibanoff et al., 2005)
  - α-maxmin (Ghirardato et al., 2004; Bleichrodt et al., 2023)
- **Main findings:**
  - For all three models, accounting for ambiguity attitudes improves predictions.
  - α-maxmin provides the best out-of-sample predictions; Smooth Ambiguity and Source Theory perform similarly.
    - Represents beliefs as a range of possible probabilities, bounded by pessimistic and optimistic beliefs.
- **Methodology:** Dynamically Optimized Sequential Experimentation (DOSE; Chapman et al., 2025a)
  - DOSE adaptively selects questions based on participants' previous responses.
  - We extend DOSE to efficiently elicit beliefs and ambiguity attitudes.

Related Literature and Contributions

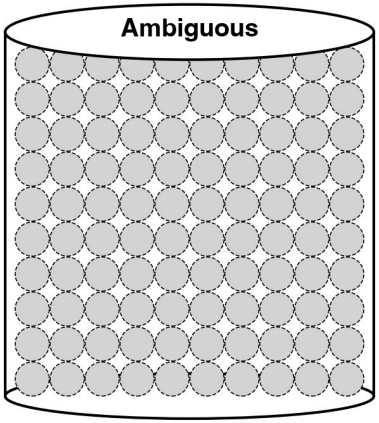
1. **Compare ambiguity attitudes across situations within a single model**
   Ert & Trautmann (2014); Baillon et al. (2018); Klingebiel & Zhu (2023); Abdellaoui et al. (2025)
2. **Compare ambiguity models within a single situation**
   Hey et al. (2010); Conte & Hey (2013); Kothiyal et al. (2014); Georgalos (2019)

→ This paper: Compare ambiguity models by testing the transferability of estimated ambiguity attitudes across situations.

Challenge: Conventional elicitation methods would require too many questions.

3. **DOSE and its applications** (loss aversion, risk preferences, time preferences, etc.)
   Chapman et al. (2025a, 2025b); Clay et al. (2016); Imai & Camerer (2018)
   → This paper: Extend DOSE to efficiently elicit beliefs and ambiguity attitudes.

How Do Models Evaluate an Ambiguous Bet?



Ambiguous bet:

One ball is drawn,

if red: \$$x$,

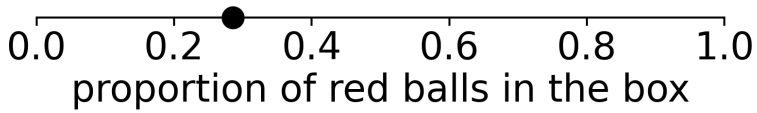
if not red: \$0.

- Beliefs: what the decision maker thinks the composition may be.
- Ambiguity attitudes: individual preferences toward ambiguity, beyond beliefs.

The three models differ in how they represent beliefs and how ambiguity attitudes affect the evaluation of this bet.

Source Theory (Abdellaoui et al., 2011; Baillon et al., 2025)

Belief: $P(\text{Red})$, a single subjective probability of drawing a red ball from the box.

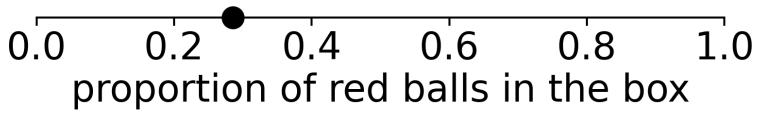


Utility of the ambiguous bet (**under expected utility**):

$$P(\text{Red})u(x) + [1 - P(\text{Red})]u(0)$$

Source Theory (Abdellaoui et al., 2011; Baillon et al., 2025)

Belief: $P(\text{Red})$, a single subjective probability of drawing a red ball from the box.



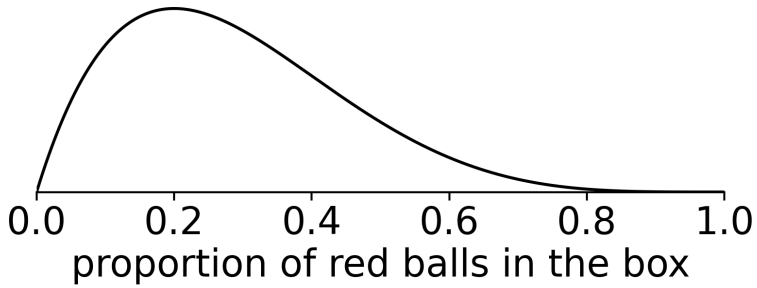
Utility of the ambiguous bet (**with ambiguity attitudes**):

$$w(P(\text{Red}))u(x) + [1 - w(P(\text{Red}))]u(0)$$

Ambiguity attitude: $w(\cdot)$, a weighting function that transforms the subjective probability when evaluating the bet.

Smooth Ambiguity (Klibanoff et al., 2005)

Belief: Π, a distribution over possible proportions p of red balls in the box.

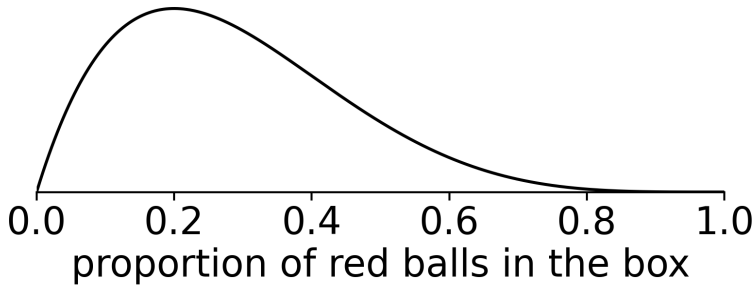


Utility of the ambiguous bet (**under expected utility**):

$$\int_0^1 pu(x) + (1-p)u(0)\, d\Pi(p)$$

Smooth Ambiguity (Klibanoff et al., 2005)

Belief: Π, a distribution over possible proportions p of red balls in the box.



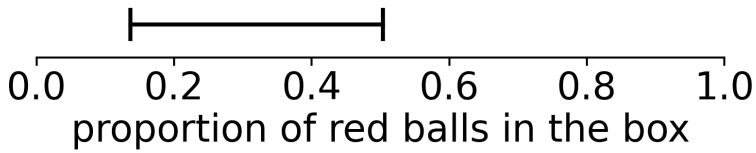
Utility of the ambiguous bet (**with ambiguity attitudes**):

$$\varphi^{-1}\left(\int_0^1 \varphi\big(pu(x)+(1-p)u(0)\big)\, d\Pi(p)\right)$$

Ambiguity attitude: $\varphi(\cdot)$ determines how utilities across possible proportions are aggregated.

α-maxmin (Ghirardato et al., 2004)

Belief: the proportion of red balls lies between $\underline{p}_{\text{Red}}$ (pessimistic belief) and $\overline{p}_{\text{Red}}$ (optimistic belief).



Utility of the ambiguous bet (**with ambiguity attitudes**):

$$\alpha\left[\underline{p}_{\text{Red}}u(x)+(1-\underline{p}_{\text{Red}})u(0)\right]+(1-\alpha)\left[\overline{p}_{\text{Red}}u(x)+(1-\overline{p}_{\text{Red}})u(0)\right]$$

Ambiguity attitude: α, weight placed on the pessimistic assessment.

α-maxmin: Linking Perceived Ambiguity to Beliefs

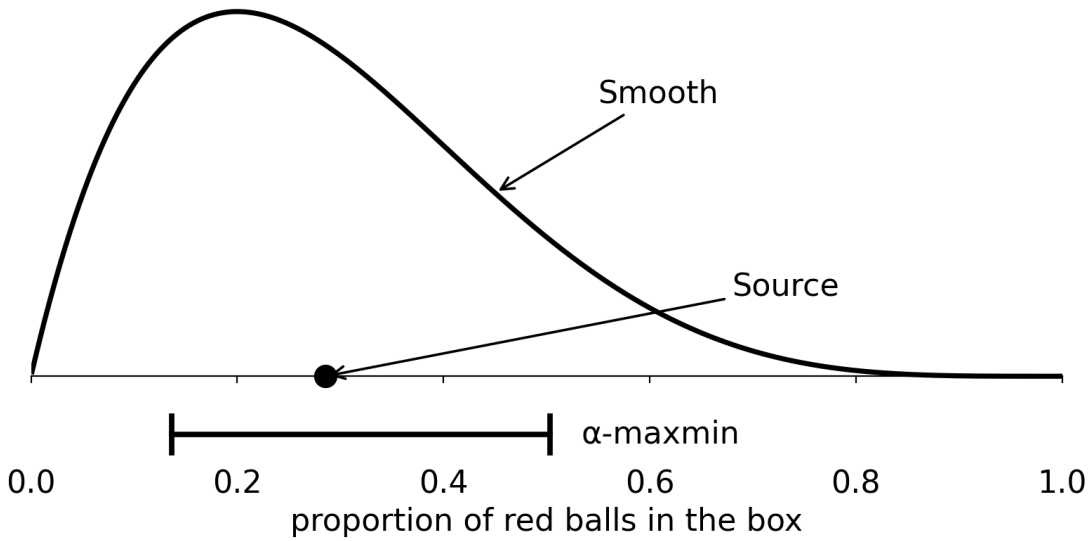
- We use the Hurwicz expected utility form (Gul & Pesendorfer, 2015; Bleichrodt et al., 2023).
- $\underline{p}_{\text{Red}}$ and $\overline{p}_{\text{Red}}$ are constructed from $P(\text{Red})$ and a parameter β.
- β determines the gap between the bounds:

$$\text{logit}(\overline{p}_{\text{Red}}) - \text{logit}(\underline{p}_{\text{Red}}) = 2 \ln\left(\frac{1+\beta}{1-\beta}\right)$$

  Naturally, β captures perceived ambiguity: a larger β generates a wider gap between the pessimistic and optimistic beliefs.
- Perceived ambiguity can change with the available information, so β should not be treated as a stable preference parameter.
- Instead, we link β to s, a measure of uncertainty in the elicited beliefs, through an ambiguity-attitude parameter κ:

$$\ln\left(\frac{1+\beta}{1-\beta}\right) = \underbrace{\kappa}_{\text{attitude}} \times \underbrace{s}_{\text{belief}}$$

Belief Representation in Each Model



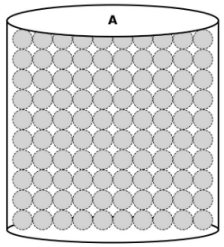
- Compared with Source Theory, Smooth Ambiguity and α-maxmin capture an additional aspect of beliefs: perceived ambiguity.
- Perceived ambiguity can change across situations.

$\rightarrow$ Source Theory may attribute effects of perceived ambiguity to ambiguity attitudes, so its ambiguity-attitude estimates may be less transferable across situations.

Experimental Design

- Subjects face two ambiguous situations: Box A (4 observed draws) and Box B (20 observed draws).
- For each box, subjects answer two types of binary-choice questions:
  - Belief questions.
  - Ambiguity questions.

Example: Belief Question



This question is about the number of blue balls in Box A.
Among 4 example draws from Box A:

Red × 2 Blue × 1 Yellow × 1

OPTION 1

Win $5 if the number of blue balls in Box A (out of 100) is:

(i) at least 0, AND

(ii) no more than 28.

Otherwise, you win $0.

CHOOSE OPTION 1

OPTION 2

Win $5 if the number of blue balls in Box A (out of 100) is:

(i) at least 29, AND

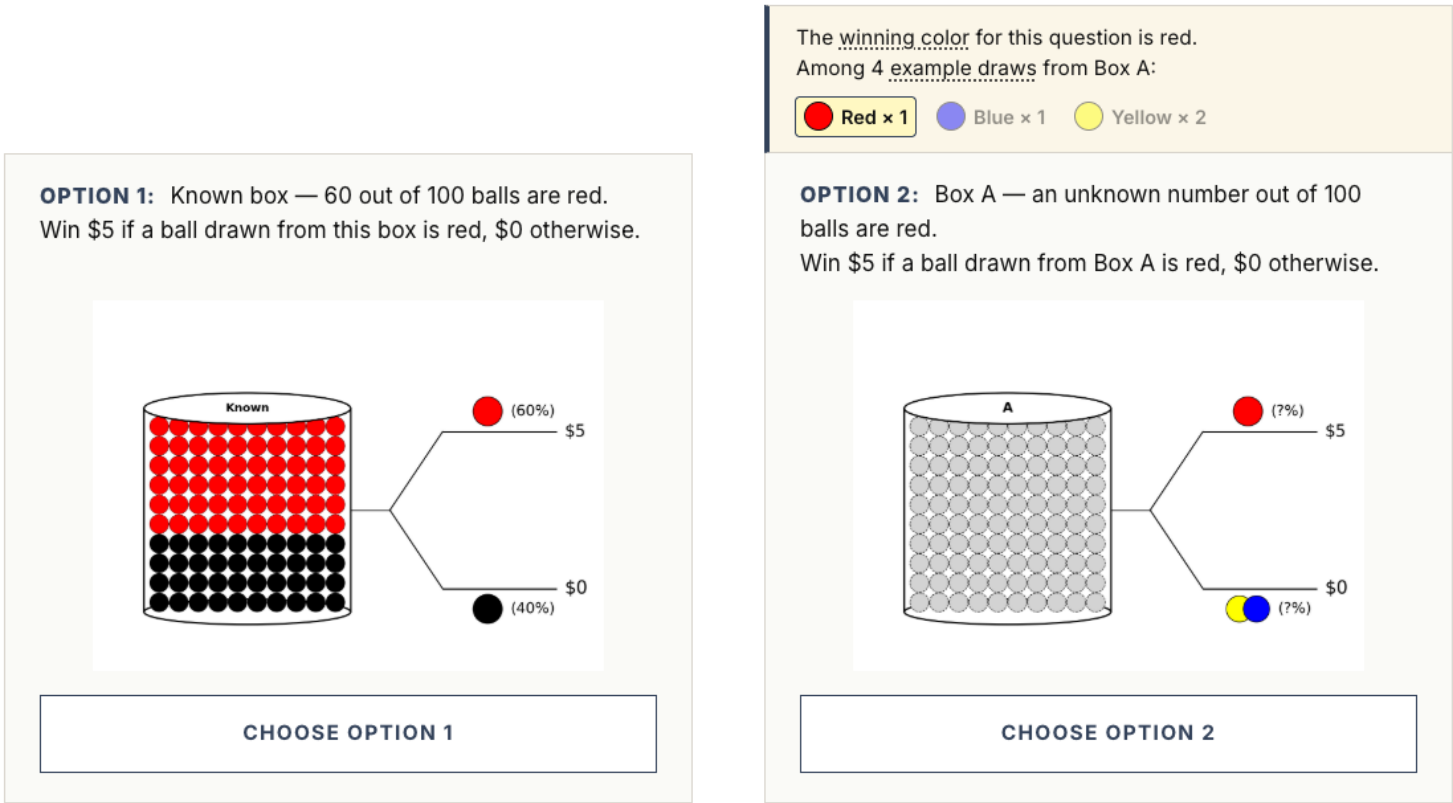
(ii) no more than 100.

Otherwise, you win $0.

CHOOSE OPTION 2

- Choices depend only on beliefs, not ambiguity-attitude parameters.
- Questions can differ in the target color and the ranges offered in each option.

Example: Ambiguity Question



- Choices depend on both belief and ambiguity-attitude parameters.
- Questions can differ in the known probability in Option 1 and the ambiguous event in Option 2.

▸ Matching probability ▸ Choice model

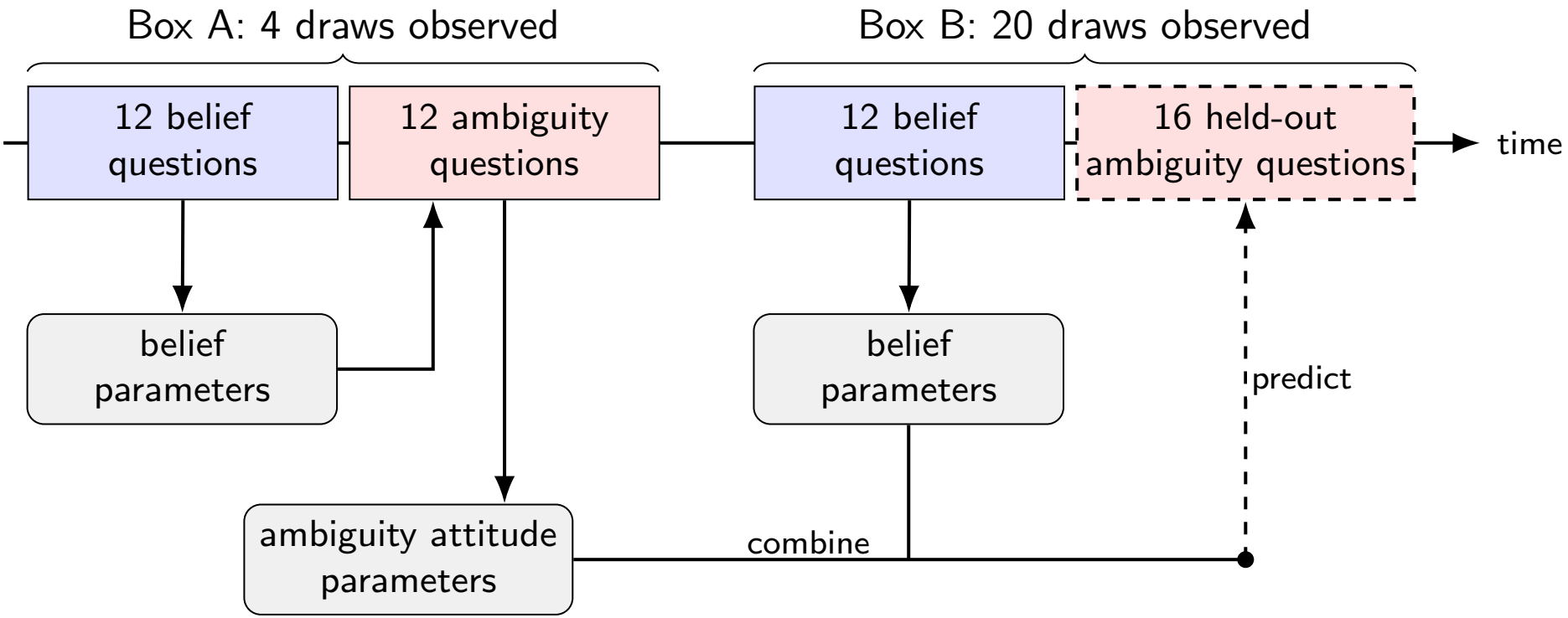
DOSE

Dynamically Optimized Sequential Experimentation (Chapman et al., 2025a)

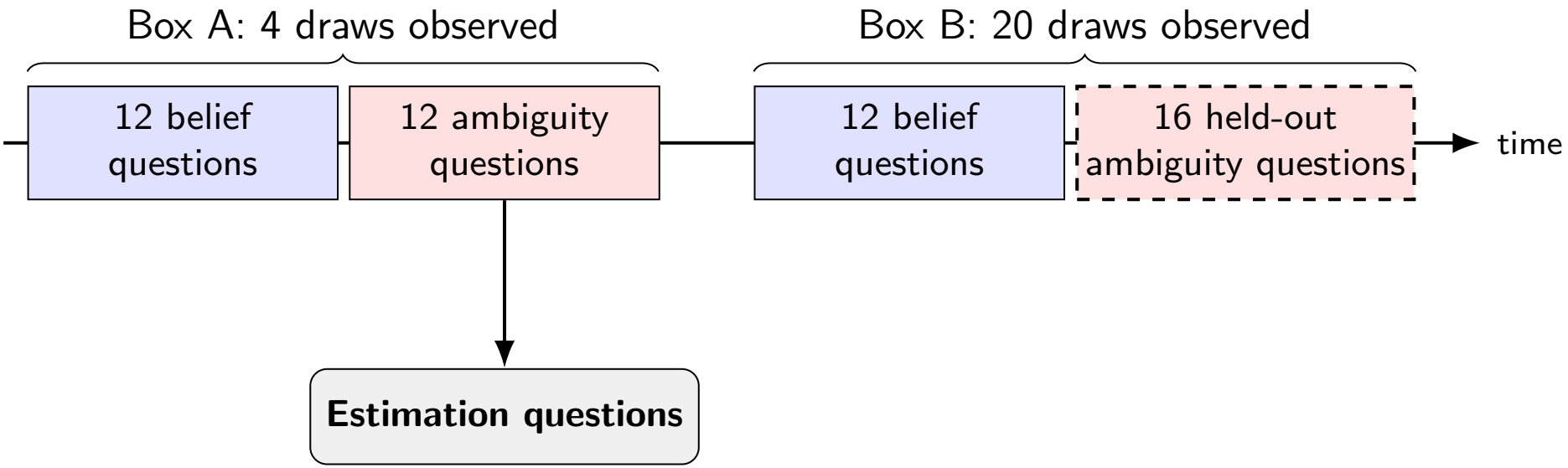
- **Challenge:** Conventional methods require too many questions to elicit both beliefs and ambiguity attitudes.
- We use DOSE to efficiently elicit beliefs and ambiguity attitudes:
  1. DOSE assumes a specific model is correct.
  2. Given this model, it evaluates how much information each candidate question is expected to provide about a subject's parameters.
  3. It asks the question expected to be most informative, updates its parameter estimates after each response, and repeats.
- **Validation:** In a separate experiment, we compare DOSE with bisection, a conventional adaptive elicitation procedure.

▸ Example ▸ Steps

Experimental Design: Overview

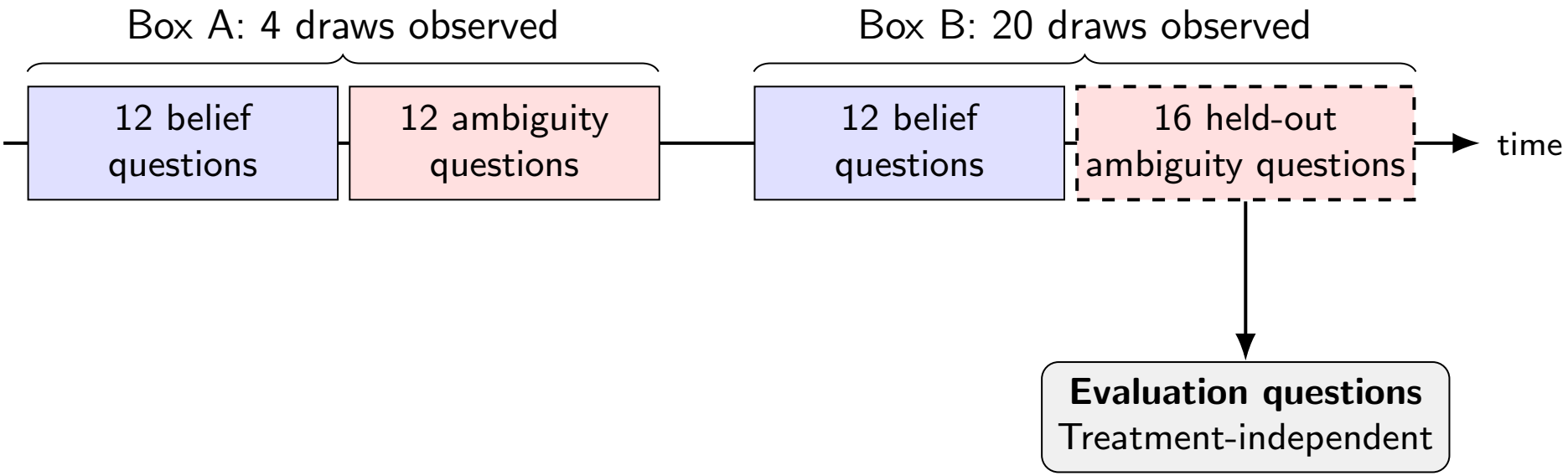


Experimental Design: Overview



- DOSE selects estimation questions based on an assumed ambiguity model.
- To ensure a fair comparison, we use three treatments, one for each ambiguity model: Source Theory, Smooth Ambiguity, and α-maxmin.

Experimental Design: Overview



- To avoid favoring any model, held-out questions are selected without relying on a specific ambiguity model.

▸ What each model reads ▸ Payment ▸ Held-out questions

Data

- Online experiment on Prolific.
- 364 subjects: 117 Source, 119 Smooth, and 128 α-maxmin.
- Median duration: 22.2 minutes.
- Payment: \$3 base payment + a possible \$5 bonus (average \$5.87).
- Validation:
  - Elicited beliefs closely track Bayesian benchmarks based on observed draws.
  - Subjects become less uncertain about the underlying probabilities after observing more draws, as reflected in their elicited beliefs ($p < 0.001$).

▸ Payment ▸ Belief validation

How We Compare the Models

We allow for stochastic choice, so each model predicts the probability of choosing each option.

Prediction score (Selten, 1998): For each held-out question, if a model assigns probability p to the option actually chosen, its score is

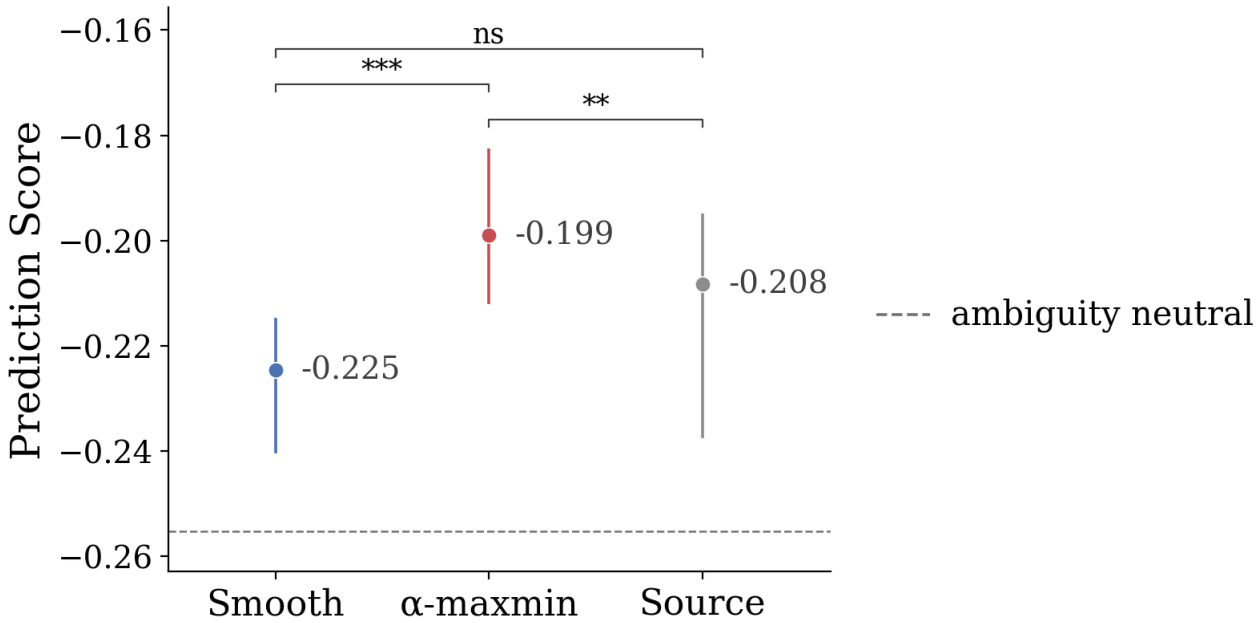
$$-(1-p)^2.$$

Average across the 16 held-out questions; **higher is better**.

▸ Outcome variable

α-maxmin Provides the Best Out-of-sample Predictions

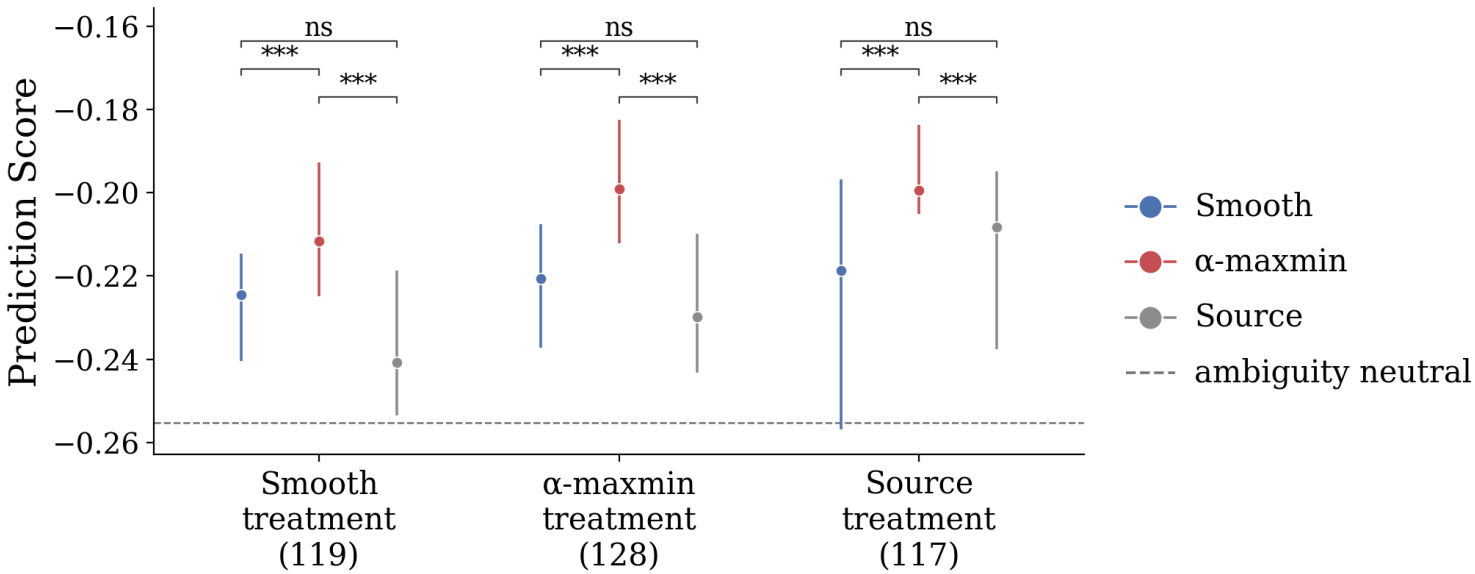
Primary analysis (pre-registered): Between-treatment comparison. Each model is evaluated using data from its corresponding DOSE treatment.



- For all three models, accounting for ambiguity attitudes improves out-of-sample predictions.
- α-maxmin significantly outperforms both alternatives; Smooth Ambiguity and Source Theory do not differ significantly.

α-maxmin's Advantage Is Robust to Question Selection

Secondary analysis (pre-registered): Within-subject comparison. We evaluate all three models using the same subjects' data within each DOSE treatment.



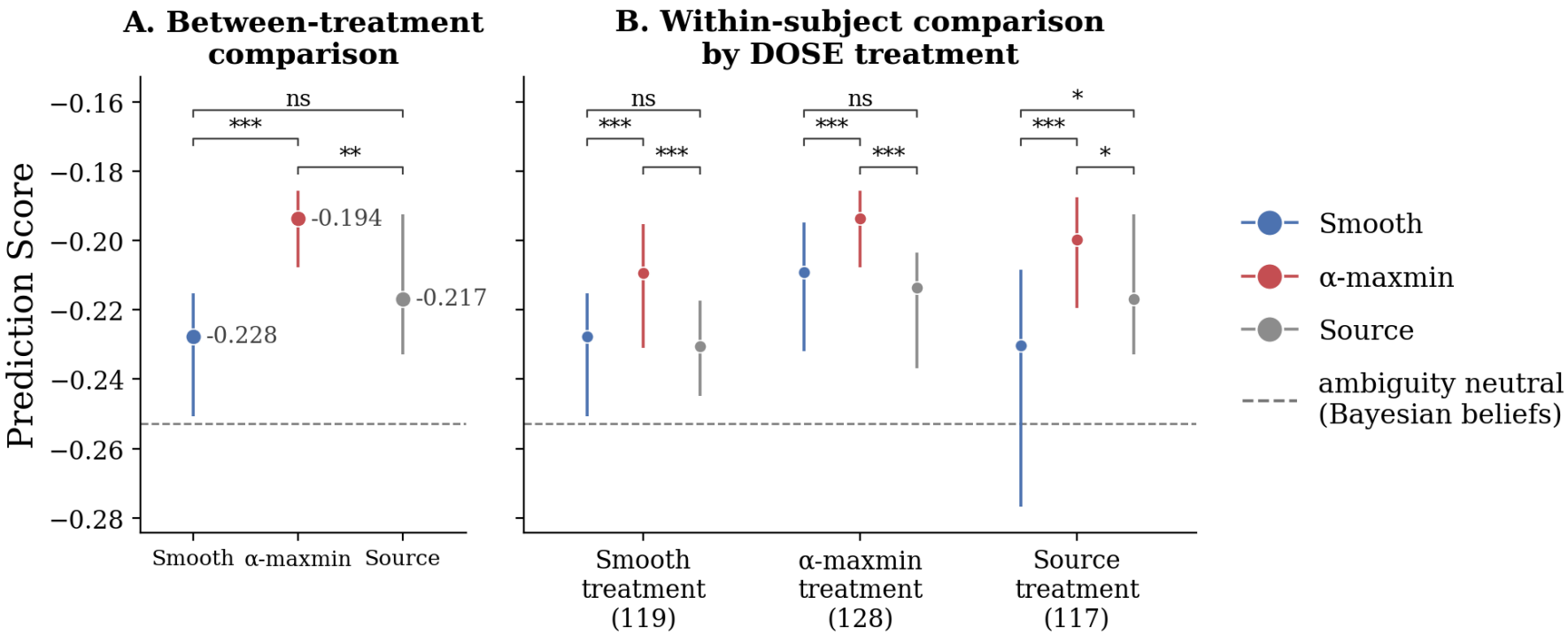
- α-maxmin performs best regardless of which DOSE treatment's data are used.
- Its advantage is not driven by differences in DOSE-selected estimation questions.

▸ In-sample fit

What If Beliefs Cannot Be Elicited?

- Our previous analysis uses beliefs elicited in both situations.
- In practice, eliciting people's beliefs may not be feasible.
- With approximated beliefs, does accounting for ambiguity attitudes still improve out-of-sample predictions, and does the same model still perform best?
- We approximate beliefs using Bayesian updating based on observed draws.

α-maxmin Still Predicts Best with Approximated Beliefs



- α-maxmin still predicts best in both between-treatment and within-subject comparisons.
- α-maxmin's advantage does not depend on directly eliciting beliefs.

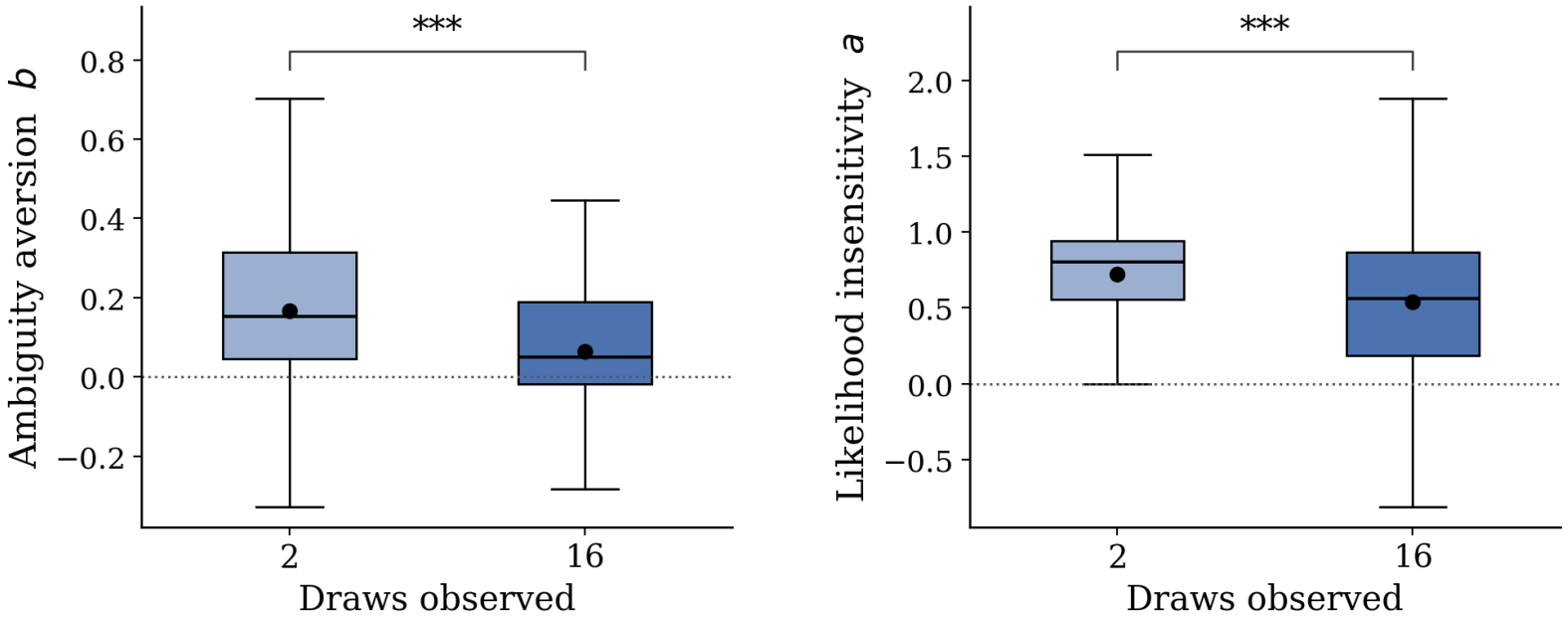
▸ Bayesian Beliefs ▸ α-maxmin Rarely Worst

Supporting Experiment 1: Benchmark

- Online experiment on Prolific, pre-registered; $N = 251$.
- Each subject faces two boxes: 2 draws observed from one box and 16 from the other.
- 24 ambiguity questions per box, using a conventional elicitation method (bisection).
- We can directly estimate ambiguity attitudes under Source Theory, but not under Smooth Ambiguity or α-maxmin, which require additional belief information.

▸ Belief representation

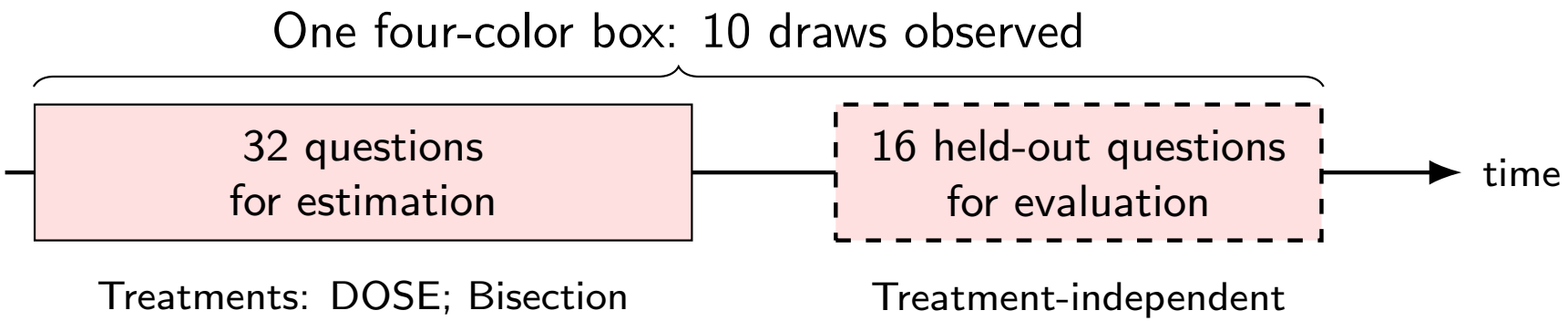
Source: More Draws → More Neutral Ambiguity Attitudes



- With more draws, estimated ambiguity attitudes under Source Theory move toward neutrality.
- More draws should reduce perceived ambiguity.
- Smooth Ambiguity and α-maxmin can explain the same behavioral pattern without any change in ambiguity attitudes.

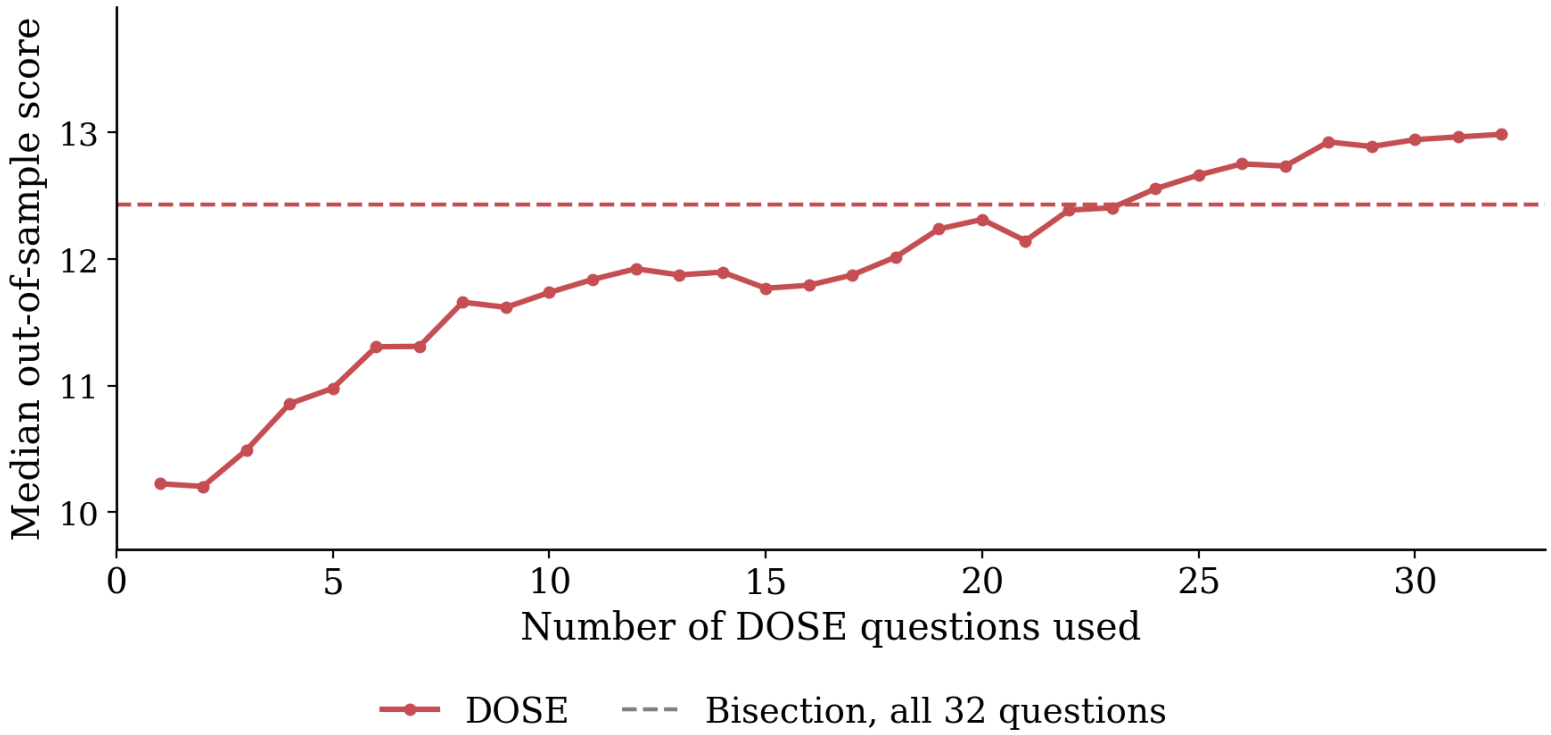
Supporting Experiment 2: DOSE Validation

- Pre-registered online experiment on Prolific ($N = 236$).



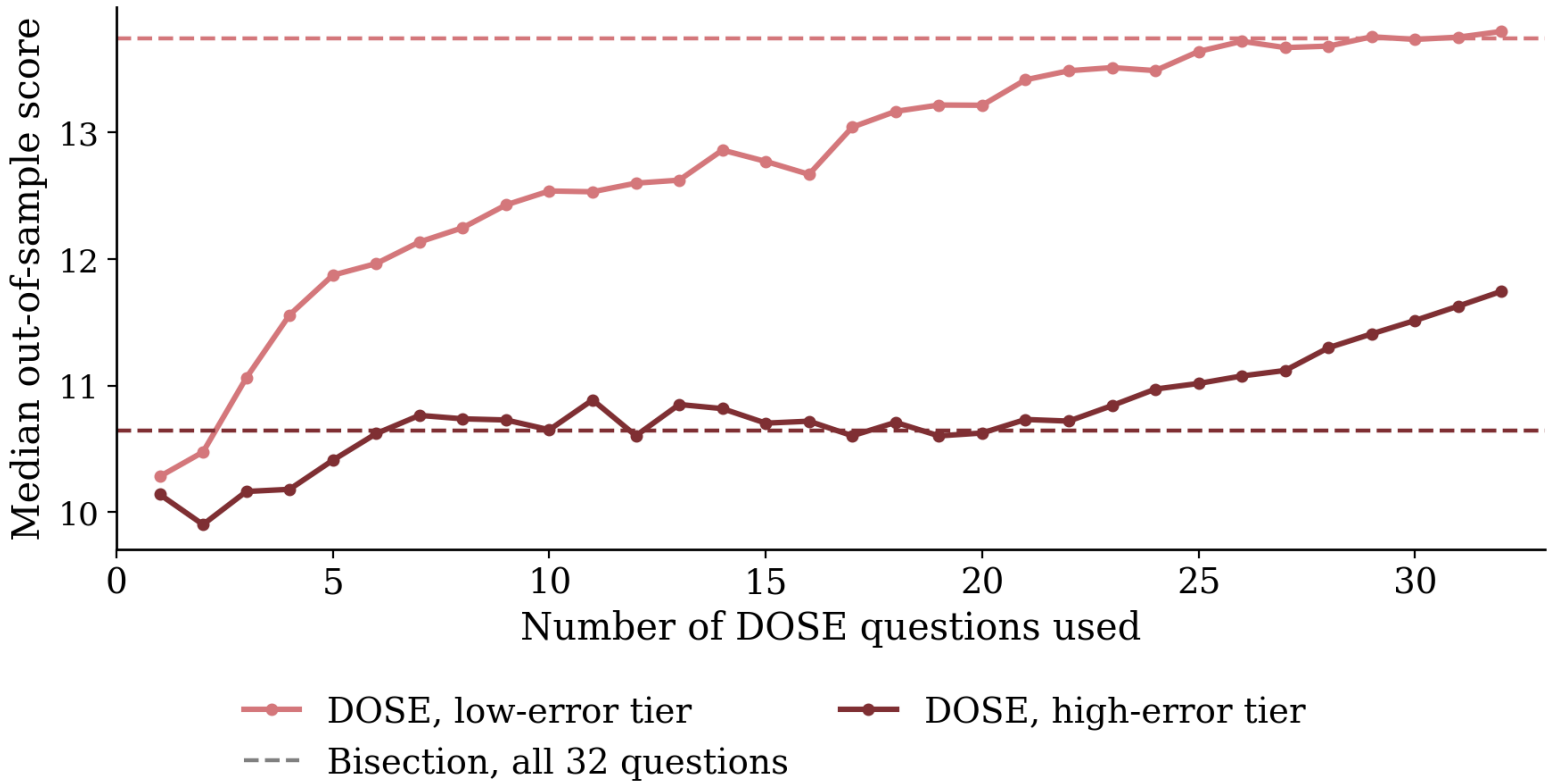
- Estimation and evaluation questions come from separate question pools.
- DOSE provides more precise parameter estimates and better held-out predictions than bisection with the same number of questions.

DOSE Matches Bisection with Fewer Questions



- DOSE reaches bisection's full 32-question score after 24 questions.

DOSE Matches Bisection with Fewer Questions



- DOSE's efficiency advantage is greater for subjects who tend to make inconsistent choices: it matches bisection's 32-question performance after just 7 questions.

Conclusion

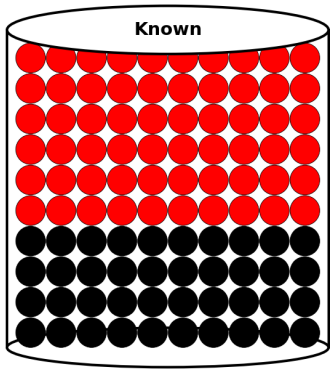
- Accounting for ambiguity attitudes improves predictions of behavior in new situations across all three models.
- α-maxmin provides the best out-of-sample predictions; Smooth Ambiguity and Source Theory perform similarly.
  - This ranking holds with both elicited and approximated beliefs.
  - This suggests that representing beliefs as a range of possible probabilities, bounded by pessimistic and optimistic beliefs, may be more appropriate under ambiguity.
  - Potentially generalizable implication: Ambiguity attitudes estimated under α-maxmin may be more transferable across situations.
- We extend DOSE to elicit beliefs and ambiguity attitudes.
  - Large efficiency gains in ambiguity attitude parameter elicitation.
  - DOSE does not appear to bias results in subsequent tasks.

Thank you!

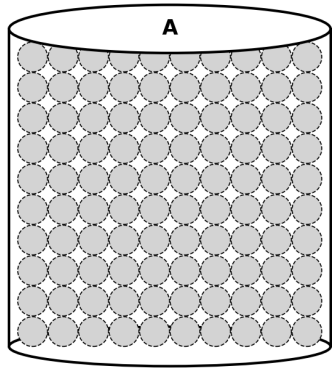
Comments or questions: teresasijia@tamu.edu

Measuring Behavior under Ambiguity

Option 1: Known box, 60 out of 100 balls are red. Win \$5 if the ball drawn is red.



Option 2: Box A, an unknown number out of 100 balls are red. Win \$5 if the ball drawn is red.



- Matching probability: The known probability that makes the subject indifferent between the two options. ▸ How each model predicts
- Why this comparison? Under appropriate assumptions:
  - The trade-off reduces to comparing the known probability in Option 1 with the matching probability of the ambiguous event in Option 2.
  - Monetary utility and risk attitudes do not directly enter this trade-off; only beliefs and ambiguity-attitude parameters matter.

◂ Back

How Each Model Predicts Matching Probability

Source Theory

$$MP_E = w(P(E))$$

Smooth Ambiguity

$$MP_E = \varphi^{-1}\left(\int_0^1 \varphi(p)\, d\Pi(p)\right)$$

α-maxmin

$$MP_E = \alpha\, \underline{p}_\kappa(P(E), s) + (1-\alpha)\, \overline{p}_\kappa(P(E), s)$$

Back

Choice Model

Belief question

$$P(\text{Option 1}) = \Phi\left(\frac{\Pi(\text{Option 1's range}) - \Pi(\text{Option 2's range})}{\sigma}\right)$$

Π captures the subject's belief distribution; σ captures choice noise.

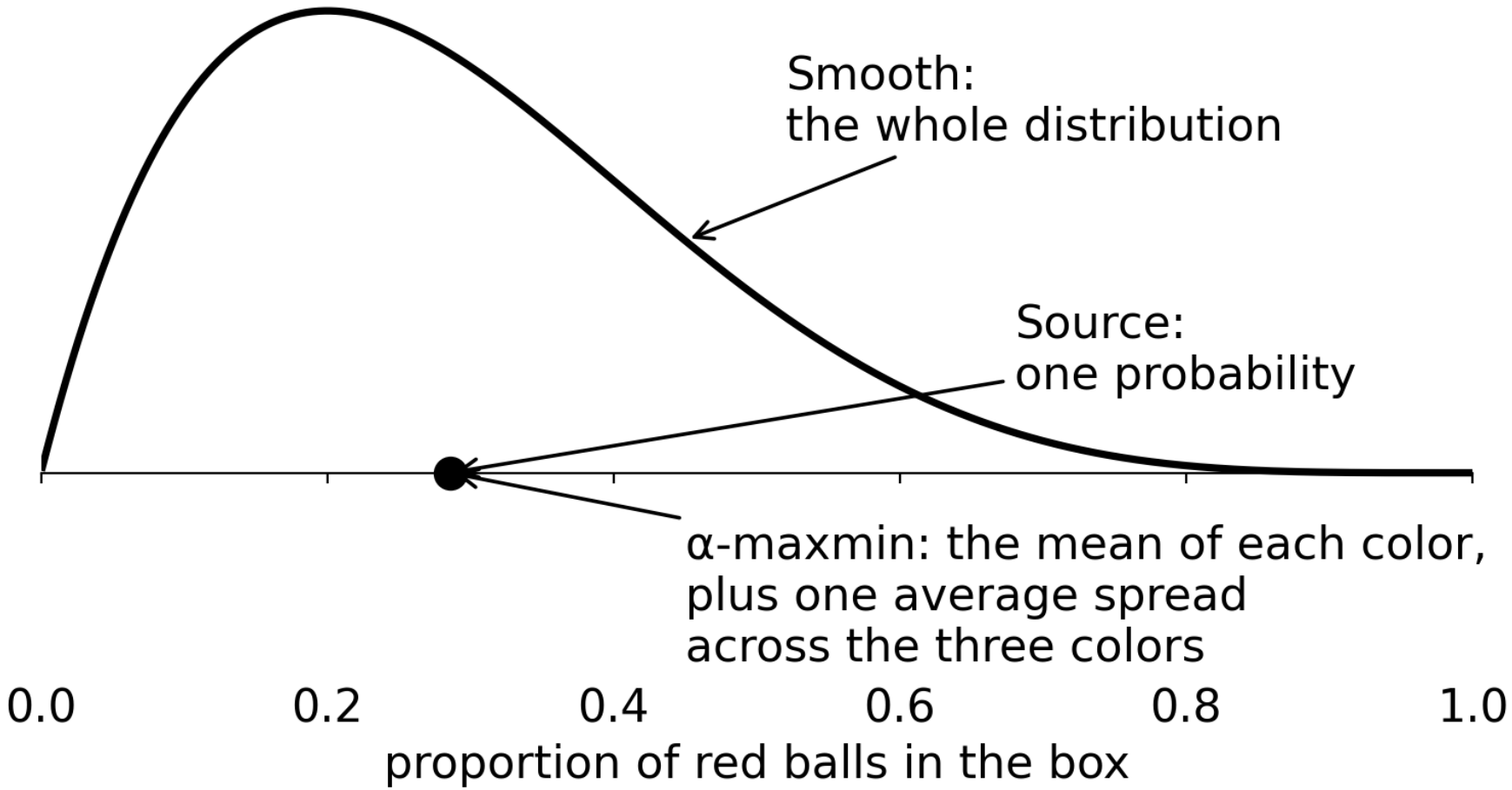
Ambiguity question

$$P(\text{Option 2}) = \Phi\left(\frac{MP_E - q}{\sigma}\right)$$

MP_E is the matching probability predicted by the model; q is the probability offered in Option 1 (the known box); σ captures choice noise.

Back

What Each Model Reads from the Belief



Back

Outcome Variable: Out-of-sample Prediction Score

- Each question asks the subject to choose between a risky bet that wins with known probability q and an ambiguous bet that wins if event E occurs.
- We model the probability of choosing the ambiguous bet as:

$$P(\text{Choose ambiguous bet}) = \Phi\left(\frac{MP_E - q}{\sigma}\right)$$

MP_E: matching probability of event E; σ: a measure of choice inconsistency (higher values indicate less consistent choices).
When $q = MP_E$, each bet is chosen with probability 0.5.

- Prediction score: If the model assigns probability p to the subject's actual choice, it scores $-(1-p)^2$, averaged over the 16 held-out questions. Higher is better.

Back

Payment

- \$3 base payment + a possible \$5 bonus.
- Before answering any questions, one question is randomly selected from the full question pool (26,610 possible questions) for payment.
  - The selection is precommitted using a code shown to subjects, without revealing the selected question.
- If the selected question is asked during the experiment, the subject's response determines payment; otherwise, it is asked at the end.
- Subjects have no incentive to strategically answer questions to influence which questions are asked later.

◂ Back to overview ◂ Back to Data

Treatment-independent Held-out Questions

- Each question asks subjects to choose between a risky bet with known probability q and an ambiguous bet on event E.
- For each ambiguous event E, we ask a series of questions that keep E fixed while adaptively varying the known probability q to locate the subject's indifference point.
- Question selection does not assume any particular ambiguity model.
- Simulations compare 2×8, 3×5, and 4×4 (events $\times$ questions per event). The 2×8 design best distinguishes the models' predictions.

Back

Example: How DOSE Selects a Question

Consider two equally likely candidate parameter values, θ_1 and θ_2, and two possible questions, Q_A and Q_B.

	Predicted probability of choosing Option 1	
	Under parameter θ_1	Under parameter θ_2
Q_A	80%	80%
Q_B	95%	5%

- Q_A: Both parameter values predict the same choice probability—not informative about the parameters.
- Q_B: The parameter values predict very different choice probabilities—more informative about the parameters.

DOSE selects Q_B because it is expected to reduce uncertainty about the parameters more.

Back

How DOSE Selects the Next Question

Before each question: a question pool and a weight $w_t(\psi)$ on each candidate parameter combination ψ, given the first t answers.

Step 1: Measure uncertainty about the parameters.

$$H(w_t) = -\sum_{\psi} w_t(\psi) \log w_t(\psi)$$

where:

- $H(w_t)$: uncertainty about the parameters.

Back

How DOSE Selects the Next Question

Step 2: Choose the question with the largest expected reduction in uncertainty.

$$Q_{t+1} = \arg\max_{Q} \left\{ \underbrace{H(w_t)}_{\substack{\text{current} \\ \text{uncertainty}}} - \underbrace{\sum_{a=1}^{2} \bar{P}_t(a \mid Q) H(w_{t+1}^{(a,Q)})}_{\substack{\text{expected uncertainty} \\ \text{after the answer}}} \right\}$$

where:

- Q: a candidate question from the question pool.
- $\bar{P}_t(a \mid Q)$: the predicted probability of choosing Option a.
- $H(w_{t+1}^{(a,Q)})$: the remaining uncertainty conditional on Option a being chosen.

Back

How DOSE Selects the Next Question

Step 3: Ask, update, and repeat.

- Ask question Q_{t+1}.
- Update the weights using Bayes' rule.
- Repeat from Step 1.

Back

Appendix: Belief Representation and Bayesian Counterpart

- **Belief representation.** We use the belief questions to estimate a Dirichlet belief over the box composition, with mean shares $\mathbf{m}$ and concentration K. For an event E, the corresponding marginal belief is

$$p_E \sim \mathrm{Beta}\big(m_E K, (1 - m_E)K\big).$$

  Each model uses a different feature of this belief.

- **Bayesian counterpart.** Instead of the belief questions, the decision maker starts from a flat prior, $\mathrm{Dirichlet}(1, 1, 1)$, and updates on the observed draws:

$$\mathrm{Dirichlet}(1 + n_r,\, 1 + n_b,\, 1 + n_y), \qquad m_c = \frac{1 + n_c}{3 + n}, \quad K = 3 + n,$$

  where n_c is the number of draws of color c and n the number of draws.

Back

Appendix: Source

Belief. Source uses the mean m_E of the belief about event E.

Attitude. A Prelec source function with parameters α_S, β_S:

$$f(m) = \exp[-\beta_S(-\ln m)^{\alpha_S}].$$

Matching probability.

$$MP_E = f(m_E).$$

Results are robust to alternative source functions, including neo-additive and Goldstein–Einhorn; see the paper appendix.

Back

Appendix: Smooth Ambiguity Model

Belief. Smooth uses the full distribution of p_E:

$$\Pi = \mathrm{Beta}\big(m_E K, (1 - m_E)K\big).$$

Attitude. An exponential-power function with parameters θ, ρ:

$$\varphi(x) = \frac{1 - e^{-\theta x^{\rho}}}{1 - e^{-\theta}}.$$

Matching probability.

$$MP_E = \varphi^{-1}\left(\int \varphi(p)\, d\Pi(p)\right).$$

Back

Appendix: α-maxmin Specification

Belief. α-maxmin uses the mean m_E and a measure of belief spread s for each box. Under the Dirichlet representation, s is the average standard deviation of the three color shares on the log-odds scale:

$$s = \frac{1}{3}\sum_c s_c, \qquad s_c^2 = \psi'(m_c K) + \psi'((1-m_c)K).$$

Ambiguity attitudes. κ captures sensitivity to belief spread, while α determines the weight on the pessimistic evaluation.

Pessimistic and optimistic bounds.

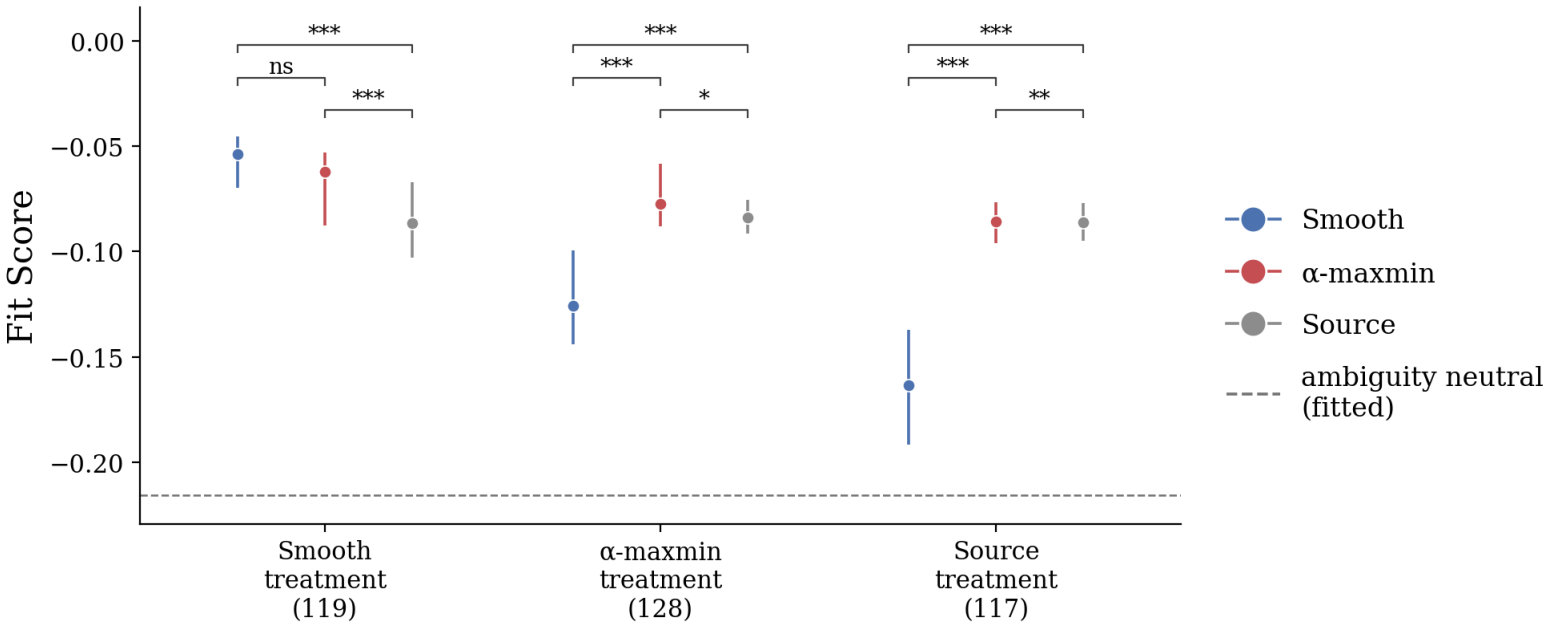
$$\text{logit}(\underline{p}_E) = \text{logit}(m_E) - \kappa s,$$
$$\text{logit}(\overline{p}_E) = \text{logit}(m_E) + \kappa s.$$

Matching probability.

$$MP_E = \alpha\, \underline{p}_E + (1-\alpha)\, \overline{p}_E.$$

Back

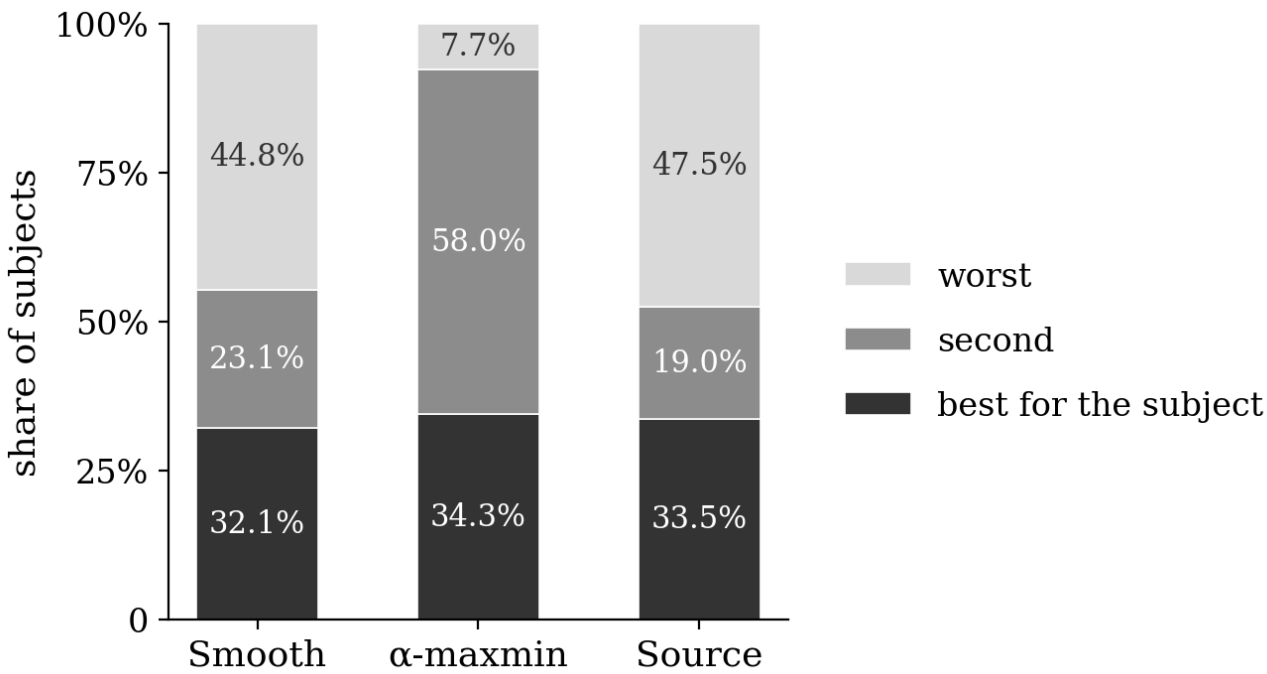
In-sample Fit Favors the Model Used for Question Selection



- Within each DOSE treatment, we compare the three models' fit on the estimation questions.
- The model used to select DOSE questions generally achieves the best in-sample fit.

Back

α-maxmin Is Rarely the Worst



- For each subject, rank the three models by their out-of-sample prediction scores.
- Each bar shows the share of subjects for whom the corresponding model ranks best, second, or worst.

Back

Elicited Beliefs Respond to the Observed Draws

Beliefs track the draws. Regress the best-guess proportion for each color on its Bayesian counterpart from the observed draws (uniform prior):

	Box A, 4 draws	Box B, 20 draws
Bayesian counterpart	0.935^{***}	0.818^{***}
	(0.033)	(0.023)
R^2	0.650	0.706
Subjects (3 colors each)	364	364

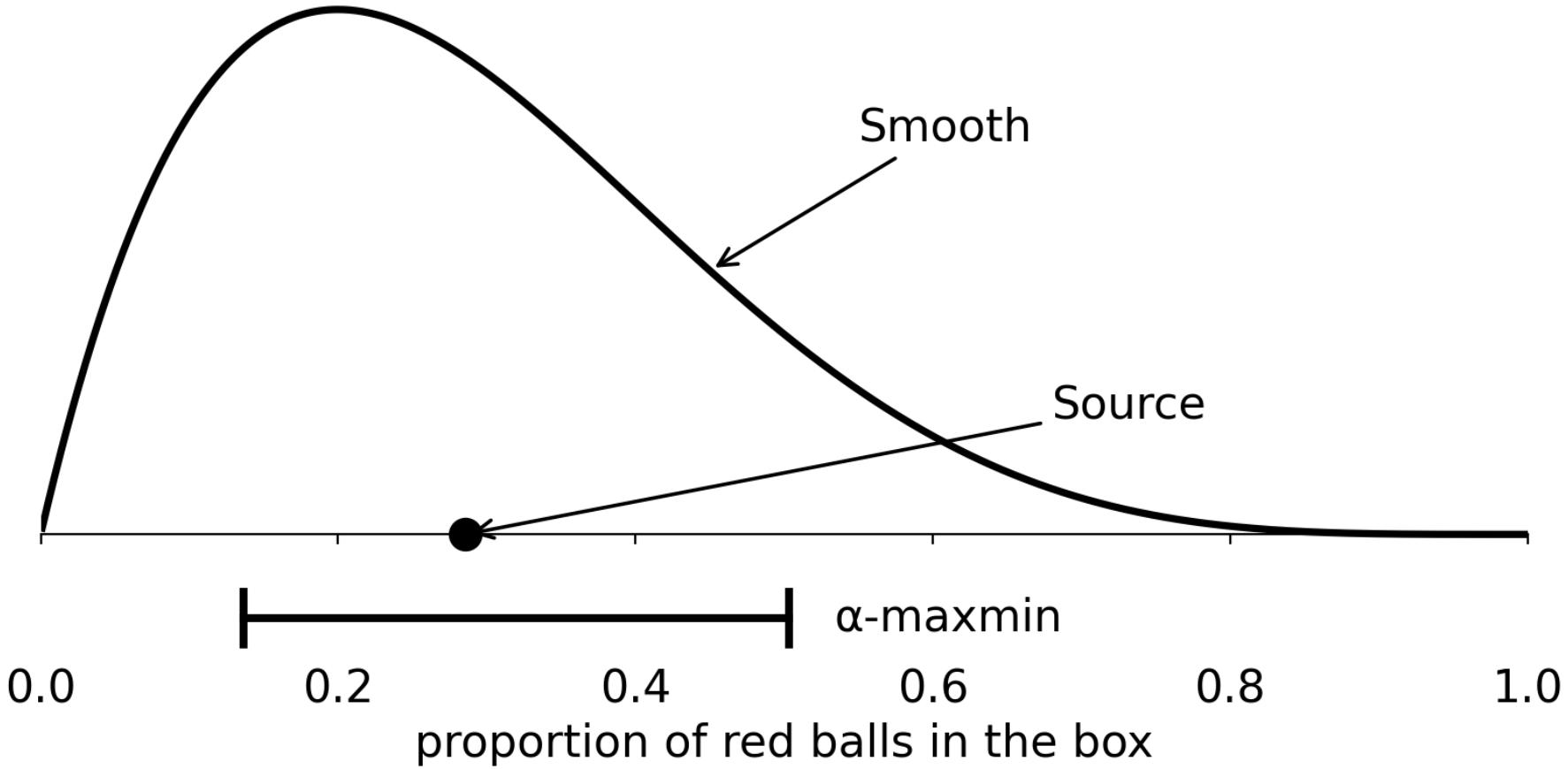
Standard errors clustered by subject. $^{***}p < 0.01$.

Perceived ambiguity falls with more draws. Perceived-ambiguity index $\mho^2$ (Izhakian, 2017, 2020):

- Median 0.0140 in Box A vs. 0.0092 in Box B (Wilcoxon signed-rank, $p < 0.001$).

Back

Belief Representation in Each Model



Back